Phishing Risk Simulation

Prepared for:
Security Awareness Program Stakeholders

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Executive Summary

Introduction

This report covers the outcomes of a phishing susceptibility simulation conducted for 12 weeks across 100 employees of a simulated organization. This project demonstrates the application of behavioral science principles to cybersecurity human risk modeling.

Objective

The goal of this simulation was to model how individual personality traits, stress levels, and phishing email characteristics influence employee click-through behavior, and to evaluate how security awareness training affects susceptibility over time.

Methodology

An agent-based model was developed in Python. Each of the 100 simulated employees was assigned two Big Five personality traits - conscientiousness and risk tolerance. Simulated employees were also given fluctuating stress levels influenced by phishing email urgency and authority characteristics. In addition, three email types (generic, urgent and authority-based) were issued across 12 weeks and training was only triggered when an employee clicked on a phishing email. Lastly, a SOC detection layer was simulated to operate independently of the behavioral model. These variables were chosen as an attempt to closely match a real-world environment.

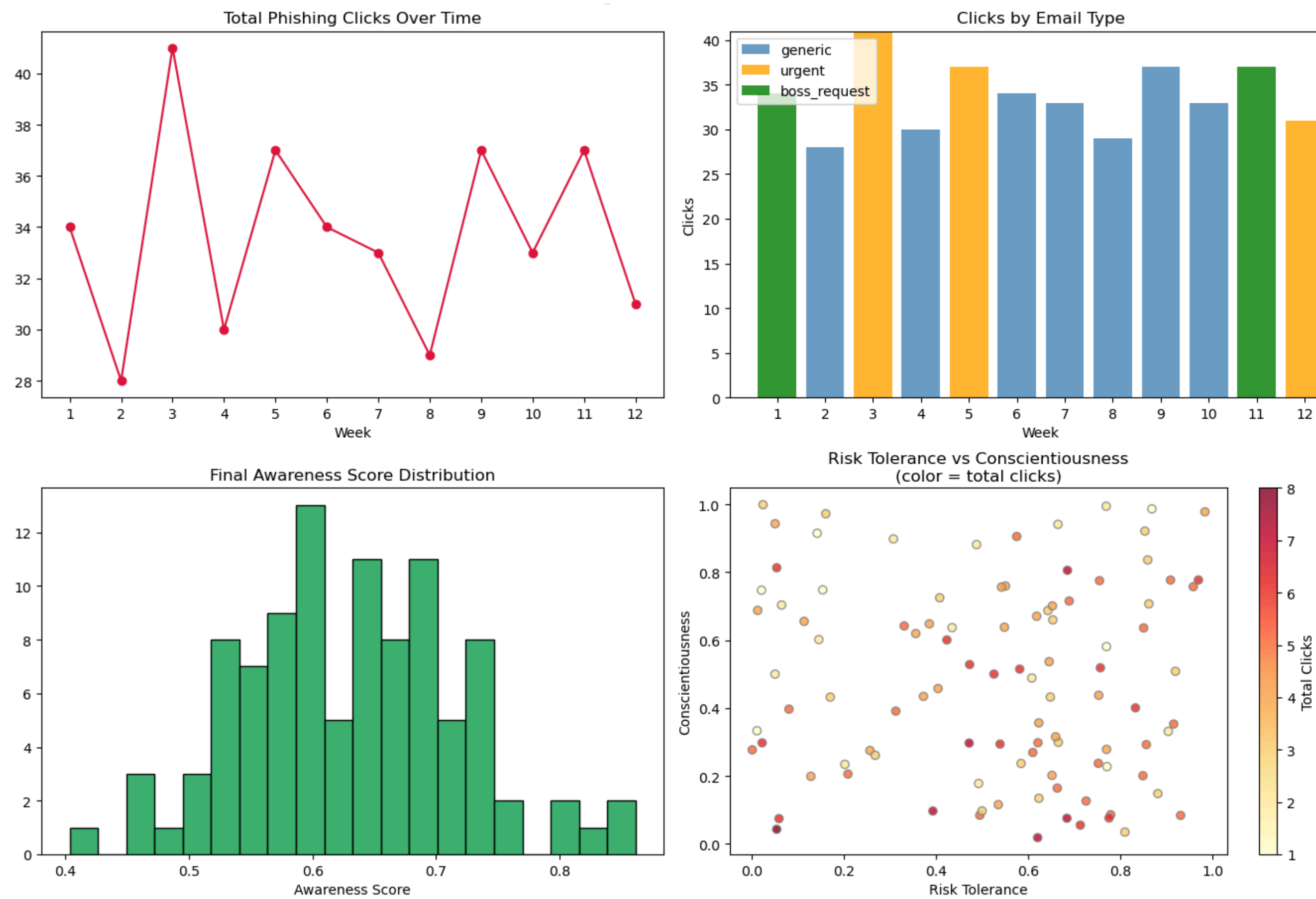
Findings

Over the course of the 12 weeks, phishing clicks ranged from 28 to 41 out of 100 employees. The clicks peaked during urgent and authority-based (boss request) email weeks . Employees with low conscientiousness and high risk tolerance clicked more often than those with high conscientiousness and low risk tolerance. Awareness decay between training events resulted in increased click rates during Weeks 9 and 11 despite an overall downward trend.
(See Table 1 & 2 below).

Table 1: Sample Run Data

Week	Email Type	Clicks	SOC Detected	High-Risk	Low-Conscientious
1	boss_request	34	22	10	17
2	generic	28	10	10	15
3	urgent	41	28	14	19
4	generic	30	14	9	11
5	urgent	37	26	16	15
6	generic	34	11	5	17
7	generic	33	16	9	14
8	generic	29	13	10	10
9	generic	37	12	9	14
10	generic	33	19	9	10
11	boss_request	37	23	11	19
12	urgent	31	18	10	11

Table 2: Phishing Simulation Results



Recommendations

- Awareness training should be prioritized for those with high risk-tolerance and low conscientiousness behavioral traits.
- To reduce awareness decay, training frequency should be increased instead of depending upon one-off and sporadic programs.
- Controls regarding authority-based and urgent emails should be implemented. These tended to influence the most clicks.
- Incorporate behavior based segmentation into current security awareness training programs in order to account for employee differences - instead of using cookie cutter methods.

Conclusion

This simulation illustrates that phishing susceptibility is influenced by various factors beyond technical controls alone. Variables such as personality, stress and authority characteristics can influence variances in phishing email clicking behavior. This simulation highlights the significance of applying Industrial Organizational (IO) psychology concepts to cybersecurity human risk training programs.